

597

(Exercise 6.3)

Q1 Check which of the following define linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^2$?

(i) $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$

Sol: Given transformation is

$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$

Let $u_1 = (x_1, x_2, x_3)$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$

(i) then we prove $T(u_1 + u_2) = T(u_1) + T(u_2)$

$$\begin{aligned} \text{Now } T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= ((x_1 + y_1) - (x_2 + y_2), (x_1 + y_1) - (x_3 + y_3)) \\ &= (x_1 + y_1 - x_2 - y_2, x_1 + y_1 - x_3 - y_3) \\ &= (x_1 - x_2 + y_1 - y_2, x_1 - x_3 + y_1 - y_3) \\ &= (x_1 - x_2, x_1 - x_3) + (y_1 - y_2, y_1 - y_3) \\ &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= T(u_1) + T(u_2) \end{aligned}$$

(ii) Let $a \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$

then we prove $T(au_1) = aT(u_1)$

$$\begin{aligned} \text{Now } T(au_1) &= T(a(x_1, x_2, x_3)) \\ &= T(ax_1, ax_2, ax_3) \\ &= (ax_1 - ax_2, ax_1 - ax_3) \\ &= a(x_1 - x_2, x_1 - x_3) \\ &= aT(x_1, x_2, x_3) \\ &= aT(u_1) \end{aligned}$$

Hence, T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$

$$(ii) T(x_1, x_2, x_3) = (|x_1|, x_2 - x_3)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (|x_1|, x_2 - x_3)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\& u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ then we prove}$$

$$(i) T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= (|x_1 + y_1|, (x_2 + y_2) - (x_3 + y_3)) \end{aligned}$$

$$\therefore T(u_1 + u_2) = (|x_1 + y_1|, x_2 + y_2 - x_3 - y_3) \quad \text{--- (1)}$$

Now

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= (|x_1|, x_2 - x_3) + (|y_1|, y_2 - y_3) \\ &= (|x_1| + |y_1|, x_2 - x_3 + y_2 - y_3) \end{aligned}$$

$$\therefore T(u_1) + T(u_2) = (|x_1| + |y_1|, x_2 + y_2 - x_3 - y_3) \quad \text{--- (2)}$$

From (1) & (2)

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$(iii) T(x_1, x_2, x_3) = (x_1 + 1, x_2 + x_3)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_1 + 1, x_2 + x_3)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\& u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ then we prove}$$

$$(i) T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$T(u_1 + u_2) = T((x_1, x_2, x_3) + (y_1, y_2, y_3))$$

$$\begin{aligned}
 T(u_1 + u_2) &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
 &= (x_1 + y_1 + 1, x_2 + y_2 + x_3 + y_3) \\
 T(u_1 + u_2) &= (x_1 + y_1 + 1, x_2 + x_3 + y_2 + y_3) \quad \text{--- (1)}
 \end{aligned}$$

Now

$$\begin{aligned}
 T(u_1) + T(u_2) &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\
 &= (x_1 + 1, x_2 + x_3) + (y_1 + 1, y_2 + y_3) \\
 &= (x_1 + 1 + y_1 + 1, x_2 + x_3 + y_2 + y_3) \\
 T(u_1) + T(u_2) &= (x_1 + y_1 + 2, x_2 + x_3 + y_2 + y_3) \quad \text{--- (2)}
 \end{aligned}$$

From (1) & (2)

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$(iv) \quad T(x_1, x_2, x_3) = (0, x_3)$$

Sol. Given transformation is

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$$T(x_1, x_2, x_3) = (0, x_3)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\text{ \& } u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ then we prove.}$$

$$(i) \quad T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned}
 T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\
 &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
 &= (0, x_3 + y_3) \\
 &= (0, x_3) + (0, y_3) \\
 &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3)
 \end{aligned}$$

$$\therefore T(u_1 + u_2) = T(u_1) + T(u_2)$$

$$(ii) \quad \text{Let } a \in \mathbb{R} \text{ \& } u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3 \text{ then we prove}$$

$$T(au_1) = aT(u_1)$$

$$\text{Now } T(au_1) = T(a(x_1, x_2, x_3))$$

$$\begin{aligned}
 T(au_1) &= T(ax_1, ax_2, ax_3) \\
 &= (0, ax_3) \\
 &= a(0, x_3) \\
 &= aT(x_1, x_2, x_3)
 \end{aligned}$$

$$T(au_1) = aT(u_1)$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$(v) \quad T(x_1, x_2, x_3) = \left(\frac{x_1 + x_2}{x_3}, x_3 \right)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = \left(\frac{x_1 + x_2}{x_3}, x_3 \right)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\text{\& } u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ then we prove}$$

$$(i) \quad T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned}
 T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\
 &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
 &= \left(\frac{x_1 + y_1 + x_2 + y_2}{x_3 + y_3}, x_3 + y_3 \right) \quad \text{--- (1)}
 \end{aligned}$$

Now

$$\begin{aligned}
 T(u_1) + T(u_2) &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\
 &= \left(\frac{x_1 + x_2}{x_3}, x_3 \right) + \left(\frac{y_1 + y_2}{y_3}, y_3 \right) \\
 &= \left(\frac{x_1 + x_2}{x_3} + \frac{y_1 + y_2}{y_3}, x_3 + y_3 \right) \\
 &= \left(\frac{y_3(x_1 + x_2) + x_3(y_1 + y_2)}{x_3 y_3}, x_3 + y_3 \right) \quad \text{--- (2)}
 \end{aligned}$$

From (1) & (2)

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$(vi) T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_3 - 3x_2 - 2x_1)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_3 - 3x_2 - 2x_1)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$ then we prove

$$(i) T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= (3(x_1 + y_1) - 2(x_2 + y_2) + (x_3 + y_3), (x_3 + y_3) - 3(x_2 + y_2) - 2(x_1 + y_1)) \\ &= (3x_1 - 2x_2 + x_3 + 3y_1 - 2y_2 + y_3, x_3 - 3x_2 - 2x_1 + y_3 - 3y_2 - 2y_1) \\ &= (3x_1 - 2x_2 + x_3, x_3 - 3x_2 - 2x_1) + (3y_1 - 2y_2 + y_3, y_3 - 3y_2 - 2y_1) \\ &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= T(u_1) + T(u_2) \end{aligned}$$

(ii) Let $\alpha \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$ then we prove

$$T(\alpha u_1) = \alpha T(u_1)$$

$$\begin{aligned} \text{Now } T(\alpha u_1) &= T(\alpha(x_1, x_2, x_3)) \\ &= T(\alpha x_1, \alpha x_2, \alpha x_3) \\ &= (3\alpha x_1 - 2\alpha x_2 + \alpha x_3, \alpha x_3 - 3\alpha x_2 - 2\alpha x_1) \\ &= \alpha(3x_1 - 2x_2 + x_3, x_3 - 3x_2 - 2x_1) \\ &= \alpha T(x_1, x_2, x_3) \\ &= \alpha T(u_1) \end{aligned}$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

Q2 Show that each of the following defines linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

$$(i) T(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_3, x_1)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_3, x_1)$$

Let $u_1 = (x_1, x_2, x_3)$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= ((x_1 + y_1) - (x_2 + y_2), (x_2 + y_2) - (x_3 + y_3), x_1 + y_1) \\ &= (x_1 + y_1 - x_2 - y_2, x_2 + y_2 - x_3 - y_3, x_1 + y_1) \\ &= (x_1 - x_2 + y_1 - y_2, x_2 - x_3 + y_2 - y_3, x_1 + y_1) \\ &= (x_1 - x_2, x_2 - x_3, x_1) + (y_1 - y_2, y_2 - y_3, y_1) \\ &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= T(u_1) + T(u_2) \end{aligned}$$

Now

$$\begin{aligned} T(au_1) &= T(a(x_1, x_2, x_3)) \\ &= T(ax_1, ax_2, ax_3) \\ &= (ax_1 - ax_2, ax_2 - ax_3, ax_1) \\ &= a(x_1 - x_2, x_2 - x_3, x_1) \\ &= aT(x_1, x_2, x_3) \\ &= aT(u_1) \end{aligned}$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$

(ii) $T(x_1, x_2, x_3) = (x_1 + x_2, -x_1 - x_2, x_3)$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_1 + x_2, -x_1 - x_2, x_3)$$

Let $u_1 = (x_1, x_2, x_3)$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned}
 T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\
 &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
 &= (x_1 + y_1 + x_2 + y_2, -(x_1 + y_1) - (x_2 + y_2), x_3 + y_3) \\
 &= (x_1 + y_1 + x_2 + y_2, -x_1 - y_1 - x_2 - y_2, x_3 + y_3) \\
 &= (x_1 + x_2 + y_1 + y_2, -x_1 - x_2 - y_1 - y_2, x_3 + y_3) \\
 &= (x_1 + x_2, -x_1 - x_2, x_3) + (y_1 + y_2, -y_1 - y_2, y_3) \\
 &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\
 &= T(u_1) + T(u_2)
 \end{aligned}$$

(ii) Let $a \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$ then we prove

$$T(au_1) = aT(u_1)$$

Now

$$\begin{aligned}
 T(au_1) &= T(a(x_1, x_2, x_3)) \\
 &= T(ax_1, ax_2, ax_3) \\
 &= (ax_1 + ax_2, -ax_1 - ax_2, ax_3) \\
 &= a(x_1 + x_2, -x_1 - x_2, x_3) \\
 &= aT(x_1, x_2, x_3) \\
 &= aT(u_1)
 \end{aligned}$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$

$$(ii) \quad T(x_1, x_2, x_3) = (x_2, -x_1, -x_3)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_2, -x_1, -x_3)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\& \text{ } u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ then we prove}$$

$$(i) \quad T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$T(u_1 + u_2) = T((x_1, x_2, x_3) + (y_1, y_2, y_3))$$

$$\begin{aligned}
T(u_1 + u_2) &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
&= (x_2 + y_2, -(x_1 + y_1), -(x_3 + y_3)) \\
&= (x_2 + y_2, -x_1 - y_1, -x_3 - y_3) \\
&= (x_2, -x_1, -x_3) + (y_2, -y_1, -y_3) \\
&= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\
&= T(u_1) + T(u_2)
\end{aligned}$$

(ii) Let $\alpha \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$ then we prove

$$T(\alpha u_1) = \alpha T(u_1)$$

Now

$$\begin{aligned}
T(\alpha u_1) &= T(\alpha(x_1, x_2, x_3)) \\
&= T(\alpha x_1, \alpha x_2, \alpha x_3) \\
&= (\alpha x_2, -\alpha x_1, -\alpha x_3) \\
&= \alpha(x_2, -x_1, -x_3) \\
&= \alpha T(x_1, x_2, x_3) \\
&= \alpha T(u_1)
\end{aligned}$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

$$(iv) \quad T(x_1, x_2, x_3) = (x_1 - 3x_2 - 2x_3, x_2 - 4x_3, x_3)$$

Sol: Given transformation is

$$T(x_1, x_2, x_3) = (x_1 - 3x_2 - 2x_3, x_2 - 4x_3, x_3)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\& \quad u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \quad \text{then we prove}$$

$$(i) \quad T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned}
T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\
&= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\
&= ((x_1 + y_1) - 3(x_2 + y_2) - 2(x_3 + y_3), (x_2 + y_2) - 4(x_3 + y_3), x_3 + y_3) \\
&= (x_1 - 3x_2 - 2x_3 + y_1 - 3y_2 - 2y_3, x_2 - 4x_3 + y_2 - 4y_3, x_3 + y_3)
\end{aligned}$$

$$T(u_1 + u_2) = (x_1 - 3x_2 - 2x_3, x_2 - 4x_3, x_3) + (y_1 - 3y_2 - 2y_3, y_2 - 4y_3, y_3) \\ = T(x_1, x_2, x_3) + T(y_1, y_2, y_3)$$

$$= T(u_1) + T(u_2)$$

(ii) Let $\alpha \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$ then we prove
 $T(\alpha u_1) = \alpha T(u_1)$

Now

$$T(\alpha u_1) = T(\alpha(x_1, x_2, x_3)) \\ = T(\alpha x_1, \alpha x_2, \alpha x_3) \\ = (3\alpha x_1 - 3\alpha x_2 - 2\alpha x_3, \alpha x_2 - 4\alpha x_3, \alpha x_3) \\ = \alpha(3x_1 - 3x_2 - 2x_3, x_2 - 4x_3, x_3) \\ = \alpha T(x_1, x_2, x_3) \\ = \alpha T(u_1)$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

$$(V) \quad T(x_1, x_2, x_3) = (x_1 + x_3, x_1 - x_3, x_2)$$

Sol. Given Transformation is

$$T(x_1, x_2, x_3) = (x_1 + x_3, x_1 - x_3, x_2)$$

$$\text{Let } u_1 = (x_1, x_2, x_3)$$

$$\& \quad u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3 \quad \text{then we prove}$$

$$T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$T(u_1 + u_2) = T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ = T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ = ((x_1 + y_1) + (x_3 + y_3), (x_1 + y_1) - (x_3 + y_3), x_2 + y_2) \\ = (x_1 + x_3 + y_1 + y_3, x_1 - x_3 + y_1 - y_3, x_2 + y_2) \\ = (x_1 + x_3, x_1 - x_3, x_2) + (y_1 + y_3, y_1 - y_3, y_2) \\ = T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ = T(u_1) + T(u_2)$$

(ii) Let $a \in \mathbb{R}$ & $u_1 = (x_1, x_2, x_3) \in \mathbb{R}^3$ then we prove
 $T(au_1) = aT(u_1)$

Now

$$\begin{aligned} T(au_1) &= T(a(x_1, x_2, x_3)) \\ &= T(ax_1, ax_2, ax_3) \\ &= (ax_1 + ax_3, ax_1 - ax_3, ax_2) \\ &= a(x_1 + x_3, x_1 - x_3, x_2) \\ &= aT(x_1, x_2, x_3) \\ &= aT(u_1) \end{aligned}$$

Hence T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

Q3 Show that each of the following transformations is not linear.

(i) $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(x_1, x_2) = x_1 x_2$

Sol. Given transformation is

$$T(x_1, x_2) = x_1 x_2$$

Let $u_1 = (x_1, x_2)$

& $u_2 = (y_1, y_2) \in \mathbb{R}^2$ then we prove

$$(i) T(u_1 + u_2) = T(u_1) + T(u_2)$$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2) + (y_1, y_2)) \\ &= T(x_1 + y_1, x_2 + y_2) \\ &= (x_1 + y_1)(x_2 + y_2) \quad \text{--- (1)} \end{aligned}$$

&

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2) + T(y_1, y_2) \\ &= x_1 x_2 + y_1 y_2 \quad \text{--- (2)} \end{aligned}$$

from (1) & (2) $T(u_1 + u_2) \neq T(u_1) + T(u_2)$

Hence T is not a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}$

(ii) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1 + 1, 2x_2, x_1 + x_2)$

Sol. Given transformation is

$$T(x_1, x_2) = (x_1 + 1, 2x_2, x_1 + x_2)$$

Let $u_1 = (x_1, x_2)$

& $u_2 = (y_1, y_2) \in \mathbb{R}^2$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2) + (y_1, y_2)) \\ &= T(x_1 + y_1, x_2 + y_2) \\ &= (x_1 + y_1 + 1, 2(x_2 + y_2), (x_1 + y_1) + (x_2 + y_2)) \\ &= (x_1 + y_1 + 1, 2x_2 + 2y_2, x_1 + x_2 + y_1 + y_2) \quad \text{--- ①} \end{aligned}$$

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2) + T(y_1, y_2) \\ &= (x_1 + 1, 2x_2, x_1 + x_2) + (y_1 + 1, 2y_2, y_1 + y_2) \\ &= (x_1 + y_1 + 2, 2x_2 + 2y_2, x_1 + x_2 + y_1 + y_2) \quad \text{--- ②} \end{aligned}$$

from ① & ②

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^3$.

(iii) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2, x_3) = (|x_1|, 0)$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (|x_1|, 0)$$

Let $u_1 = (x_1, x_2, x_3)$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \end{aligned}$$

$$T(u_1 + u_2) = (|x_1 + y_1|, 0) \quad \text{————— ①}$$

4

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= (|x_1|, 0) + (|y_1|, 0) \\ &= (|x_1| + |y_1|, 0) \quad \text{————— ②} \end{aligned}$$

From ① & ②

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

(iv) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2) = (x_1^2, x_2^2)$.

Sol. Given transformation is

$$T(x_1, x_2) = (x_1^2, x_2^2)$$

Let $u_1 = (x_1, x_2)$

& $u_2 = (y_1, y_2) \in \mathbb{R}^2$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2) + (y_1, y_2)) \\ &= T(x_1 + y_1, x_2 + y_2) \\ &= ((x_1 + y_1)^2, (x_2 + y_2)^2) \quad \text{—————} \end{aligned}$$

4

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2) + T(y_1, y_2) \\ &= (x_1^2, x_2^2) + (y_1^2, y_2^2) \\ &= (x_1^2 + y_1^2, x_2^2 + y_2^2) \quad \text{————— ②} \end{aligned}$$

from ① & ②

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$.

(v) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2, x_3) = (x_1, x_2, x_3) + (1, 1, 1)$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_1, x_2, x_3) + (1, 1, 1)$$

Let $u_1 = (x_1, x_2, x_3)$

& $u_2 = (y_1, y_2, y_3) \in \mathbb{R}^3$ then we prove

(i) $T(u_1 + u_2) = T(u_1) + T(u_2)$

Now

$$\begin{aligned} T(u_1 + u_2) &= T((x_1, x_2, x_3) + (y_1, y_2, y_3)) \\ &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= (x_1 + y_1, x_2 + y_2, x_3 + y_3) + (1, 1, 1) \\ &= (x_1 + y_1 + 1, x_2 + y_2 + 1, x_3 + y_3 + 1) \quad \text{--- (1)} \end{aligned}$$

&

$$\begin{aligned} T(u_1) + T(u_2) &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3) \\ &= (x_1, x_2, x_3) + (1, 1, 1) + (y_1, y_2, y_3) + (1, 1, 1) \\ &= (x_1 + 1, x_2 + 1, x_3 + 1) + (y_1 + 1, y_2 + 1, y_3 + 1) \\ &= (x_1 + y_1 + 2, x_2 + y_2 + 2, x_3 + y_3 + 2) \quad \text{--- (2)} \end{aligned}$$

for (1) & (2)

$$T(u_1 + u_2) \neq T(u_1) + T(u_2)$$

Hence T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

Q3 Determine which of the following transformations are linear:

(a) $T: M_{22} \rightarrow \mathbb{R}$ defined by

(i) $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d$

Sol. Given transformation is

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d$$

Let $A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$

4 & $A_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in M_{22}$ then we prove

(i) $T(A_1 + A_2) = T(A_1) + T(A_2)$

Now

$$T(A_1 + A_2) = T\left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}\right)$$

$$= T\left(\begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix}\right)$$

$$= a_1 + a_2 + d_1 + d_2 \quad \text{--- (1)}$$

4

$$T(A_1) + T(A_2) = T\left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}\right) + T\left(\begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}\right)$$

$$= a_1 + d_1 + a_2 + d_2$$

$$= a_1 + a_2 + d_1 + d_2 \quad \text{--- (2)}$$

from (1) & (2)

$$T(A_1 + A_2) = T(A_1) + T(A_2)$$

(ii) Let $a \in R$ & $A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \in M_{22}$ then we prove

$$T(aA_1) = aT(A_1)$$

Now $T(aA_1) = T\left(a \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}\right)$

$$= T\left(\begin{bmatrix} aa_1 & ab_1 \\ ac_1 & ad_1 \end{bmatrix}\right)$$

$$= aa_1 + ad_1$$

$$= a(a_1 + d_1)$$

$$= aT\left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}\right)$$

$$= aT(A_1)$$

Hence T is a linear transformation from M_{22} to R

(ii) $T: M_{22} \rightarrow R$ defined by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

Sol. Given transformation is

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

Let $A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$

& $A_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$

then we prove

$$T(A_1 + A_2) = T(A_1) + T(A_2)$$

Now

$$T(A_1 + A_2) = T\left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}\right)$$

$$= T\left(\begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix}\right)$$

$$= (a_1 + a_2)(d_1 + d_2) - (b_1 + b_2)(c_1 + c_2) \quad \text{--- ①}$$

Now

$$T(A_1) + T(A_2) = T\left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}\right) + T\left(\begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}\right)$$

$$= a_1 d_1 - b_1 c_1 + a_2 d_2 - b_2 c_2 \quad \text{--- ②}$$

from ① & ②

$$T(A_1 + A_2) \neq T(A_1) + T(A_2)$$

Hence T is a linear transformation from M_{22} to R .

(b) $T: P_2(x) \rightarrow P_2(x)$ defined by

(i) $T(a + bx + cx^2) = a + (b+c)x + (2a-3b)x^2$

Sol. Given transformation is

$$T(a + bx + cx^2) = a + (b+c)x + (2a-3b)x^2$$

Let $U = a + bx + cx^2$

& $V = p + qx + rx^2 \in P_2(x)$ then we prove

$$(i) T(U+V) = T(U) + T(V)$$

Now

$$\begin{aligned} T(U+V) &= T((a+bx+cx^2) + (p+qx+rx^2)) \\ &= T((a+p) + (b+q)x + (c+r)x^2) \\ &= (a+p) + (b+q+c+r)x + (2a+2p-3b-3q)x^2 \\ &= (a+p) + (b+c+q+r)x + (2a-3b+2p-3q)x^2 \\ &= (a+(b+c)x + (2a-3b)x^2) + (p+(q+r)x + (2p-3q)x^2) \\ &= T(a+bx+cx^2) + T(p+qx+rx^2) \\ &= T(U) + T(V) \end{aligned}$$

(ii) Let $k \in R$ & $U = a+bx+cx^2$ then we prove

$$T(kU) = kT(U)$$

Now

$$\begin{aligned} T(kU) &= T(k(a+bx+cx^2)) \\ &= T(ka+kbx+kcx^2) \\ &= ka + (kb+kc)x + (2ka-3kb)x^2 \\ &= k(a + (b+c)x + (2a-3b)x^2) \\ &= kT(a+bx+cx^2) \\ &= kT(U) \end{aligned}$$

Hence T is a linear transformation from $P_2(x)$ to $P_2(x)$.

(ii) $T: P_2(x) \longrightarrow P_2(x)$ defined by

$$T(a+bx+cx^2) = (a+1) + bx + cx^2$$

Sol: Given transformation is

$$T(a+bx+cx^2) = (a+1) + bx + cx^2$$

Let $U = a+bx+cx^2$

& $V = p+qx+rx^2 \in P_2(x)$ then we prove

$$(i) T(u+v) = T(u) + T(v)$$

Now

$$\begin{aligned} T(u+v) &= T((a+bx+cx^2) + (b+vx+lx^2)) \\ &= T((a+b) + (b+v)x + (c+l)x^2) \\ &= (a+b+1) + (b+v)x + (c+l)x^2 \quad \text{--- (1)} \end{aligned}$$

+

$$\begin{aligned} T(u) + T(v) &= T(a+bx+cx^2) + T(b+vx+lx^2) \\ &= (a+1) + bx + cx^2 + (b+1) + vx + lx^2 \\ &= (a+b+2) + (b+v)x + (c+l)x^2 \quad \text{--- (2)} \end{aligned}$$

From (1) & (2)

$$T(u+v) \neq T(u) + T(v)$$

Hence T is not a linear transformation from $P_2(x)$ to $P_2(x)$.

Q5. If A is an $m \times n$ matrix, show that $T(x) = Ax$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ where

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

Sol.

Given transformation is

$$T(x) = Ax$$

Suppose

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n$$

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then we prove

$$T(x+y) = T(x) + T(y)$$

Now

$$T(x+y) =$$

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Q6. Determine which of the following linear transformations are one-to-one:

(i) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1 + 2x_2)$

Sol: Given linear transformation is

$$T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1 + 2x_2)$$

Let $x = (x_1, x_2)$

Let $y = (y_1, y_2) \in \mathbb{R}^2$

Then $T(x) = (x_1 + x_2, x_1 - x_2, x_1 + 2x_2)$

Let $T(y) = (y_1 + y_2, y_1 - y_2, y_1 + 2y_2)$

Suppose $T(x) = T(y)$

$$\Rightarrow (x_1 + x_2, x_1 - x_2, x_1 + 2x_2) = (y_1 + y_2, y_1 - y_2, y_1 + 2y_2)$$

$$\Rightarrow x_1 + x_2 = y_1 + y_2 \quad \text{--- (1)}$$

$$x_1 - x_2 = y_1 - y_2 \quad \text{--- (2)}$$

$$x_1 + 2x_2 = y_1 + 2y_2 \quad \text{--- (3)}$$

Adding (1) & (2)

$$2x_1 = 2y_1$$

$$\boxed{x_1 = y_1}$$

Put in (1)

$$x_1 + x_2 = x_1 + y_2$$

$$\Rightarrow \boxed{x_2 = y_2}$$

Hence $(x_1, x_2) = (y_1, y_2)$

or $x = y$

Hence $T(x) = T(y) \Rightarrow x = y$

Hence T is one-to-one

(ii) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$

Sol: Given linear transformation is

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$$

Let $x = (x_1, x_2, x_3)$

& $y = (y_1, y_2, y_3) \in \mathbb{R}^3$

Then $T(x) = (x_1 - x_2, x_3)$

& $T(y) = (y_1 - y_2, y_3)$

Suppose $T(x) = T(y)$

$\Rightarrow (x_1 - x_2, x_3) = (y_1 - y_2, y_3)$

$\Rightarrow x_1 - x_2 = y_1 - y_2$ ——— ①

$x_3 = y_3$ ——— ②

From ① we cannot conclude that

$x_1 = y_1$ & $x_2 = y_2$

Hence $T(x) = T(y) \nRightarrow x = y$

So T is not one-to-one.

(iii) $T: \mathbb{R}^2 \xrightarrow{3} \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1, x_1 + x_2, x_1 - x_2)$

Sol. Given linear transformation is

$T(x_1, x_2) = (x_1, x_1 + x_2, x_1 - x_2)$

Let $x = (x_1, x_2)$

& $y = (y_1, y_2) \in \mathbb{R}^2$

Then $T(x) = (x_1, x_1 + x_2, x_1 - x_2)$

& $T(y) = (y_1, y_1 + y_2, y_1 - y_2)$

Suppose $T(x) = T(y)$

$\Rightarrow (x_1, x_1 + x_2, x_1 - x_2) = (y_1, y_1 + y_2, y_1 - y_2)$

or $x_1 = y_1$ ——— ①

$x_1 + x_2 = y_1 + y_2$ ——— ②

$x_1 - x_2 = y_1 - y_2$ ——— ③

① $\Rightarrow$ $x_1 = y_1$

Subst. ② & ③

$2x_2 = 2y_2 \Rightarrow$ $x_2 = y_2$

$$\text{Hence } (x_1, x_2) = (y_1, y_2)$$

$$\text{or } x = y$$

$$\text{So } T(x) = T(y) \Rightarrow x = y$$

Hence T is one-to-one.

Q7 Let C be the vector space of Complex numbers over the field of reals & $T: C \rightarrow C$ be defined by $T(z) = \bar{z}$ where $\bar{z}$ denotes the Complex Conjugate of z . Show that T is linear.

Sol. Given transformation is

$$T(z) = \bar{z}$$

Let $z_1, z_2 \in C$ then we prove

$$(i) \quad T(z_1 + z_2) = T(z_1) + T(z_2)$$

Now

$$\begin{aligned} T(z_1 + z_2) &= \overline{z_1 + z_2} \\ &= \bar{z}_1 + \bar{z}_2 \\ &= T(z_1) + T(z_2) \end{aligned}$$

(ii) Let $\alpha \in R$ & $z_1 \in C$ then we prove

$$T(\alpha z_1) = \alpha T(z_1)$$

Now

$$\begin{aligned} T(\alpha z_1) &= \overline{\alpha z_1} \\ &= \alpha \bar{z}_1 \quad \because \alpha \in R \\ &= \alpha T(z_1) \end{aligned}$$

Hence T is a linear transformation from C to C .

Q8 Let V be the vector space $P_n(x)$ of polynomials $p(x)$ with real coefficients & of degree not exceeding n together with the zero polynomial. Let $T: V \rightarrow V$

be defined by $T(p(x)) = p(x+1)$

Show that T is linear.

Q.1 Given transformation is

$$T(p(x)) = p(x+1)$$

$$\text{Let } p_1(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$\text{+ } p_2(x) = b_0 + b_1x + b_2x^2 + \dots + b_nx^n \in V$$

$$\text{Then we prove } T(p_1(x) + p_2(x)) = T(p_1(x)) + T(p_2(x))$$

Now

$$\begin{aligned} T(p_1(x) + p_2(x)) &= T((a_0 + a_1x + \dots + a_nx^n) + (b_0 + b_1x + \dots + b_nx^n)) \\ &= T((a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n) \\ &= (a_0 + b_0) + (a_1 + b_1)(x+1) + \dots + (a_n + b_n)(x+1)^n \\ &= [a_0 + a_1(x+1) + \dots + a_n(x+1)^n] + [b_0 + b_1(x+1) + \dots + b_n(x+1)^n] \\ &= T(a_0 + a_1x + \dots + a_nx^n) + T(b_0 + b_1x + \dots + b_nx^n) \\ &= T(p_1(x)) + T(p_2(x)) \end{aligned}$$

$$(ii) \text{ Let } a \in \mathbb{R} \text{ + } p_1(x) = a_0 + a_1x + \dots + a_nx^n$$

$$\text{Then we prove } T(ap_1(x)) = aT(p_1(x))$$

Now

$$\begin{aligned} T(ap_1(x)) &= T(a(a_0 + a_1x + \dots + a_nx^n)) \\ &= T(aa_0 + aa_1x + \dots + aa_nx^n) \\ &= aa_0 + aa_1(x+1) + \dots + aa_n(x+1)^n \\ &= a(a_0 + a_1(x+1) + \dots + a_n(x+1)^n) \\ &= aT(a_0 + a_1x + \dots + a_nx^n) \\ &= aT(p_1(x)) \end{aligned}$$

Hence T is a linear transformation from V to V .

Q.2 Let $V_1 = (1, 1, 1)$, $V_2 = (1, 1, 0)$ + $V_3 = (1, 0, 0)$ be a basis for $\mathbb{R}^3$. Find a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ s.t.
 $T(V_1) = (1, 0)$, $T(V_2) = (2, -1)$ + $T(V_3) = (4, 3)$

Sol. Let $x = (x_1, x_2, x_3)$ be any vector of R^3 then
for scalars a_1, a_2, a_3

$$x = a_1 v_1 + a_2 v_2 + a_3 v_3$$

$$= a_1(1, 1, 1) + a_2(1, 1, 0) + a_3(1, 0, 0)$$

$$(x_1, x_2, x_3) = (a_1 + a_2 + a_3, a_1 + a_2, a_1)$$

$$\Rightarrow a_1 + a_2 + a_3 = x_1 \quad \text{--- (1)}$$

$$a_1 + a_2 = x_2 \quad \text{--- (2)}$$

$$a_1 = x_3 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow a_1 = x_3$$

Put in (2)

$$x_3 + a_2 = x_2 \Rightarrow a_2 = x_2 - x_3$$

Put in (1)

$$x_3 + x_2 - x_3 + a_3 = x_1$$

$$a_3 = x_1 - x_2$$

So

$$x = x_3 v_1 + (x_2 - x_3) v_2 + (x_1 - x_2) v_3$$

Applying T on both sides

$$T(x) = T(x_3 v_1 + (x_2 - x_3) v_2 + (x_1 - x_2) v_3)$$

$$= x_3 T(v_1) + (x_2 - x_3) T(v_2) + (x_1 - x_2) T(v_3) \quad (\text{As } T \text{ is linear})$$

$$= x_3(1, 0) + (x_2 - x_3)(2, -1) + (x_1 - x_2)(4, 3)$$

$$= (x_3, 0) + (2x_2 - 2x_3, -x_2 + x_3) + (4x_1 - 4x_2, 3x_1 - 3x_2)$$

$$= (x_3 + 2x_2 - 2x_3 + 4x_1 - 4x_2, -x_2 + x_3 + 3x_1 - 3x_2)$$

$$T(x_1, x_2, x_3) = (4x_1 - 2x_2 - x_3, 3x_1 - 4x_2 + x_3)$$

which is req. linear transformation from R^3 to R^2

Q10 Let $T: R^2 \rightarrow R$ be the linear transformation for which
 $T(1, 1) = 3$ & $T(0, 1) = -2$. Find $T(x_1, x_2)$

Soln. First we prove that the vectors $(1,1)$ & $(0,1)$ form a basis for $\mathbb{R}^2$.

Suppose for scalars $a, b \in \mathbb{R}$

$$a(1,1) + b(0,1) = 0$$

$$(a, a) + (0, b) = 0$$

$$(a, a+b) = 0$$

$$\Rightarrow a = 0 \quad \text{--- (1)}$$

$$a+b = 0 \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow \boxed{a = 0}$$

$$\textcircled{2} \Rightarrow \boxed{b = 0}$$

Hence vectors $(1,1)$ & $(0,1)$ are linearly independent.

As there are two linearly independent vectors in $\mathbb{R}^2$

So $(1,1)$ & $(0,1)$ form a basis for $\mathbb{R}^2$

Suppose $(x_1, x_2) \in \mathbb{R}^2$ be an arbitrary vector

$$\text{then } (x_1, x_2) = a(1,1) + b(0,1) \quad \text{where } a, b \in \mathbb{R}$$

$$a(x_1, x_2) = (a, a+b)$$

$$\Rightarrow a = x_1 \quad \text{--- (1)}$$

$$a+b = x_2 \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow \boxed{a = x_1}$$

Put in (2)

$$x_1 + b = x_2 \Rightarrow \boxed{b = x_2 - x_1}$$

So

$$(x_1, x_2) = x_1(1,1) + (x_2 - x_1)(0,1)$$

Applying T on both sides

$$T(x_1, x_2) = T(x_1(1,1) + (x_2 - x_1)(0,1))$$

$$= x_1 T(1,1) + (x_2 - x_1) T(0,1)$$

$$= x_1(3) + (x_2 - x_1)(-2)$$

$$= 3x_1 - 2x_2 + 2x_1$$

$$T(x_1, x_2) = 5x_1 - 2x_2 \quad \text{which is } T \text{ in terms of Co-ords.}$$

Q11 Let $D: P_2(x) \rightarrow P_2(x)$ be the differentiation operator
 & $D(p(x)) = p'(x)$ for all $p(x) \in P_2(x)$. Find $N(D)$.

Sol. Given operator is.

$$D(p(x)) = p'(x)$$

Here $N(D)$ will consist of those polynomials in $P_2(x)$ for which $D(p(x)) = 0$.

Since we know that

$$D(p(x)) = 0 \text{ if } p(x) = \text{const polynomial}$$

So $N(D)$ will consist of all const. polynomials.

Q12 Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$.

Find $N(T)$. Is T one-to-one?

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$$

$$\text{Here } N(T) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : T(x_1, x_2, x_3) = (0, 0, 0)\}$$

$$\text{Now } T(x_1, x_2, x_3) = (0, 0, 0)$$

$$\Rightarrow (-x_3, x_1, x_1 + x_3) = (0, 0, 0)$$

$$\Rightarrow -x_3 = 0 \quad \text{--- (1)}$$

$$x_1 = 0 \quad \text{--- (2)}$$

$$x_1 + x_3 = 0 \quad \text{--- (3)}$$

$$\text{(1)} \Rightarrow x_3 = 0$$

$$\text{(2)} \Rightarrow x_1 = 0$$

which shows that $N(T)$ will consist of all vectors of the form $(0, x_2, 0)$. which is x_2 -axis.

$$\text{i.e., } N(T) = \{(0, x_2, 0) \in \mathbb{R}^3 : x_2 \in \mathbb{R}\}$$

Since $N(T) = (0, x_2, 0) \neq (0, 0, 0)$. So T is not one-to-one.

Q13 Suppose U, V & W are vector spaces over the same field F . Let $T: U \rightarrow V$ & $S: V \rightarrow W$ be linear transformations. The transformation $S \circ T: U \rightarrow W$ is defined by $(S \circ T)(u) = S(T(u))$ for all $u \in U$. Show that $S \circ T$ is a linear transformation.

Sol.

Here $S \circ T: U \rightarrow W$ be defined as

$$(S \circ T)(u) = S(T(u)) \quad \text{for all } u \in U$$

Let $u_1, u_2 \in U$ then we prove

$$(S \circ T)(u_1 + u_2) = (S \circ T)(u_1) + (S \circ T)(u_2)$$

Now

$$(S \circ T)(u_1 + u_2) = S(T(u_1 + u_2))$$

$$= S(T(u_1) + T(u_2))$$

$$= S(T(u_1)) + S(T(u_2))$$

$$= (S \circ T)(u_1) + (S \circ T)(u_2)$$

By def. of $S \circ T$

$\because T$ is linear

$\because S$ is linear

(ii) Let $a \in F$ & $u \in U$ then we prove

$$(S \circ T)(au) = a(S \circ T)(u)$$

Now

$$(S \circ T)(au) = S(T(au))$$

$$= S(aT(u))$$

$$= aS(T(u))$$

$$= a(S \circ T)(u)$$

By def of $S \circ T$

$\because T$ is linear

$\because S$ is linear

Hence $S \circ T$ is a linear transformation from U to W .

Q14 Let U & V be two vector spaces over the same field F . Denote the set of all linear transformations from U into V by $L(U, V)$. Show that $L(U, V)$ is a vector space over F with vector space operations as defined in example 31

Sol. Consider the set $L(U, V)$. Let $S, T \in L(U, V)$

Now $S: U \rightarrow V$ & $T: U \rightarrow V$ be two linear transformations. Define

$S+T: U \rightarrow V$ & $\alpha S: U \rightarrow V$ by

$$(S+T)(u) = S(u) + T(u)$$

$$(\alpha S)(u) = \alpha S(u) \quad \text{for all } u \in U \text{ & } \alpha \in F$$

First we show that $L(U, V)$ is an abelian gr. under +.

(i) Closure law

Let $S, T \in L(U, V)$, then we show $S+T \in L(U, V)$.

$$\begin{aligned} \text{Now } (S+T)(u_1+u_2) &= S(u_1+u_2) + T(u_1+u_2) && \text{By def. of } S+T \\ &= S(u_1) + S(u_2) + T(u_1) + T(u_2) && \because S, T \text{ are linear} \\ &= S(u_1) + T(u_1) + S(u_2) + T(u_2) \\ &= (S+T)(u_1) + (S+T)(u_2) \end{aligned}$$

Let $K \in F$ & $u \in U$

$$\begin{aligned} (S+T)(Ku) &= S(Ku) + T(Ku) \\ &= KS(u) + KT(u) && \because S, T \text{ are linear} \\ &= K(S(u) + T(u)) \\ &= K(S+T)(u) \end{aligned}$$

Hence $S+T$ is linear & so $S+T \in L(U, V)$.

(ii) Associative law

Let $R, S, T \in L(U, V)$ then we prove

$$R + (S+T) = (R+S) + T$$

Now Consider for $u \in U$

$$\begin{aligned}
[R + (S + T)](u) &= R(u) + (S + T)(u) \\
&= R(u) + [S(u) + T(u)] \\
&= [R(u) + S(u)] + T(u) \\
&= [(R + S)(u)] + T(u) \\
&= [(R + S)(u) + T(u)] \\
&= [(R + S) + T](u)
\end{aligned}$$

(By def. of sum)

" " "

 $\Rightarrow R(u), S(u), T(u) \in F$

$$\Rightarrow R + (S + T) = (R + S) + T$$

So, $+$ is associative in $L(U, V)$.(iii) Identity lawClearly the zero transformation 0 defined by

$$0(u) = 0 \quad \text{for all } u \in U$$

is a linear transformation from U to V & it is the additive identity in $L(U, V)$.(iv) Inverse lawFor each $T \in L(U, V)$, we define

$$-T \in L(U, V) \text{ by}$$

$$(-T)(u) = -T(u)$$

then $-T$ is the additive inverse of T .(v) Commutative lawLet $S, T \in L(U, V)$ then we show $S + T = T + S$

Now Consider

$$\begin{aligned}
(S + T)(u) &= S(u) + T(u) \\
&= T(u) + S(u) \\
&= (T + S)(u)
\end{aligned}$$

By def. of sum

 $\Rightarrow S(u), T(u) \in F$

$$\Rightarrow S + T = T + S$$

Hence $+$ is commutative in $L(U, V)$ So, $L(U, V)$ is an abelian gr. under $+$.

Now we check scalar multiplication axioms.

(i) Let $a \in F$ & $S \in L(U, V)$ then we prove $aS \in L(U, V)$.

$$\begin{aligned} \text{Now } (aS)(u_1 + u_2) &= a[S(u_1 + u_2)] \\ &= a[S(u_1) + S(u_2)] && \Rightarrow S \text{ is linear} \\ &= aS(u_1) + aS(u_2) \end{aligned}$$

Suppose $K \in F$ & $u \in U$ then

$$\begin{aligned} (aS)(Ku) &= a[S(Ku)] \\ &= a[KS(u)] && \Rightarrow S \text{ is linear} \\ &= (aK)S(u) \\ &= (Ka)S(u) \\ &= K(aS)(u) \end{aligned}$$

Hence aS is linear & so $aS \in L(U, V)$.

(ii) Let $a, b \in F$ & $S \in L(U, V)$ then we prove $a(bS) = (ab)S$

$$\begin{aligned} \text{Now } [a(bS)](u) &= a.(bS)(u) \\ &= a[b.S(u)] \\ &= (ab).S(u) \\ &= [(ab)S](u) \end{aligned}$$

$$\Rightarrow a(bS) = (ab)S$$

(iii) Let $a, b \in F$ & $S \in L(U, V)$ then we prove $(a+b)S = aS + bS$

$$\begin{aligned} \text{Now } [(a+b)S](u) &= (a+b).S(u) \\ &= a.S(u) + b.S(u) \\ &= (aS)(u) + (bS)(u) \\ &= [aS + bS](u) \end{aligned}$$

(iv) Let $a \in F$ & $S, T \in L(U, V)$ then we prove $a(S+T) = aS + aT$

$$\begin{aligned} \text{Now } [a(S+T)](u) &= a[(S+T)(u)] \\ &= a[S(u) + T(u)] \\ &= a.S(u) + a.T(u) \\ &= (aS)(u) + (aT)(u) \end{aligned}$$

$$s. [a(s+t)](u) = [as+at](u)$$

$$\Rightarrow a(s+t) = as+at$$

(iv) Let $1 \in F$ & $S \in L(U, V)$ then we prove $1.S = S$

$$\text{Now, } (1.S)(u) = 1.S(u)$$

$$= S(u)$$

$$\Rightarrow 1.S = S$$

Since all the conditions are satisfied. $s. L(U, V)$ is a vector space over F .

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Q15 Find a basis & dimension of each of $R(T)$ & $N(T)$, where

(i) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 - x_3, x_2 + x_3, x_1 + x_2 - 2x_3)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 - x_3, x_2 + x_3, x_1 + x_2 - 2x_3)$$

Since $\mathbb{R}^3$ is generated by $(1, 0, 0)$, $(0, 1, 0)$ & $(0, 0, 1)$. So

$R(T)$ will be generated by $T(1, 0, 0)$, $T(0, 1, 0)$ & $T(0, 0, 1)$

$$\text{Here } T(1, 0, 0) = (1, 0, 1)$$

$$T(0, 1, 0) = (2, 1, 1)$$

$$\& T(0, 0, 1) = (-1, 1, -2)$$

Hence $R(T)$ is generated by $(1, 0, 1)$, $(2, 1, 1)$ & $(-1, 1, -2)$

$$\text{Since } (2, 1, 1) = 3(1, 0, 1) + 1(-1, 1, -2)$$

So Casting out the vector $(2, 1, 1)$, the set $\{(1, 0, 1), (-1, 1, -2)\}$

also spans $R(T)$. Since none of the two vectors is a multiple of other, so the set $\{(1, 0, 1), (-1, 1, -2)\}$ is linearly independent & so forms a basis for $R(T)$.

$$\text{Hence } \dim R(T) = 2$$

Now we find $\dim N(T)$.

A vector $(x_1, x_2, x_3) \in N(T)$ if $T(x_1, x_2, x_3) = 0$

$$\text{i.e., if } (x_1 + 2x_2 - x_3, x_2 + x_3, x_1 + x_2 - 2x_3) = (0, 0, 0)$$

$$\Rightarrow x_1 + 2x_2 - x_3 = 0 \quad \text{--- (1)}$$

$$x_2 + x_3 = 0 \quad \text{--- (2)}$$

$$x_1 + x_2 - 2x_3 = 0 \quad \text{--- (3)}$$

Adding (1) & (2)

$$x_1 + 3x_2 = 0$$

$$\text{or } \boxed{x_1 = -3x_2}$$

Put in (3)

$$-3x_2 + x_2 - 2x_3 = 0$$

$$-2x_2 - 2x_3 = 0$$

$$x_2 + x_3 = 0$$

$$\text{or } \boxed{x_3 = -x_2}$$

$$\text{If } x_2 = 1$$

$$\text{then } x_1 = -3, x_2 = 1, x_3 = -1$$

So the vector $(-3, 1, -1)$ spans $N(T)$. Also $(-3, 1, -1)$ is linearly independent. So $\{(-3, 1, -1)\}$ forms a basis for $N(T)$.

$$\text{Hence } \dim N(T) = 1$$

(ii) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ is defined by

$$T(x_1, x_2, x_3) = (2x_1 + x_3, 4x_1 + x_2, x_1 + x_3, x_3 - 4x_2)$$

Sol. Given transformation is

$$T(x_1, x_2, x_3) = (2x_1 + x_3, 4x_1 + x_2, x_1 + x_3, x_3 - 4x_2)$$

Since $\mathbb{R}^3$ is generated by $(1, 0, 0)$, $(0, 1, 0)$ & $(0, 0, 1)$. So $R(T)$ will be generated by $T(1, 0, 0)$, $T(0, 1, 0)$ & $T(0, 0, 1)$

$$\text{Here } T(1, 0, 0) = (2, 4, 1, 0)$$

$$T(0, 1, 0) = (0, 1, 0, -4)$$

$$T(0, 0, 1) = (1, 0, 1, 1)$$

Hence $R(T)$ will be generated by $(2, 4, 1, 0)$, $(0, 1, 0, -4)$, $(1, 0, 1, 1)$

Now we check whether these vectors are linearly independent. For this let

$$a(2, 4, 1, 0) + b(0, 1, 0, -4) + c(1, 0, 1, 1) = (0, 0, 0, 0) \quad \text{where } a, b, c \in F$$

$$\text{or } (2a + c, 4a + b, a + c, -4b + c) = (0, 0, 0, 0)$$

$$\Rightarrow 2a + c = 0 \quad \text{--- ①}$$

$$4a + b = 0 \quad \text{--- ②}$$

$$a + c = 0 \quad \text{--- ③}$$

$$-4b + c = 0 \quad \text{--- ④}$$

$$\text{①} - \text{③} \Rightarrow \boxed{a = 0}$$

$$\text{③} \Rightarrow 0 + c = 0 \Rightarrow \boxed{c = 0}$$

$$\text{②} \Rightarrow 0 + b = 0 \Rightarrow \boxed{b = 0}$$

Hence vectors $(2, 4, 1, 0), (0, 1, 0, -4) \text{ \& } (1, 0, 1, 1)$ are linearly independent. Hence $\{(2, 4, 1, 0), (0, 1, 0, -4), (1, 0, 1, 1)\}$ form a basis for $R(T)$.

$$\text{Hence } \dim R(T) = 3$$



Q16 Show that linear transformations preserve linear dependence.

Sol. Let $T: U \rightarrow V$ be a linear transformation, where U & V are vector spaces over the same field F . Suppose a set $\{u_1, u_2, \dots, u_n\}$ in U is linearly dependent. We want to show that $\{T(u_1), T(u_2), \dots, T(u_n)\}$ is a linearly dependent set in V .

Since $\{u_1, u_2, \dots, u_n\}$ is linearly dependent, so there exist scalars $a_1, a_2, \dots, a_n$, not all zero, such that

$$a_1 u_1 + a_2 u_2 + \dots + a_n u_n = 0$$

Applying T on both sides

$$T(a_1 u_1 + a_2 u_2 + \dots + a_n u_n) = T(0)$$

$$\text{or } a_1 T(u_1) + a_2 T(u_2) + \dots + a_n T(u_n) = 0 \quad (\because T \text{ is linear})$$

Since $a_1, a_2, \dots, a_n$ are not all zero, so the above eq. shows that $\{T(u_1), T(u_2), \dots, T(u_n)\}$ are linearly dependent in V . Hence T preserve linear dependence.

Q17 Find the rank of each matrix in problem 8 of exercise 3.2 by the method of 6.42

(i)
$$\begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 5 & -1 \\ -2 & 3 \end{bmatrix}$$

Sol.

Let $A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 5 & -1 \\ -2 & 3 \end{bmatrix}$

Then

$$\text{rank } A = 1 + \text{rank} \begin{bmatrix} \begin{vmatrix} 1 & 3 \\ 0 & -2 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ 5 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ -2 & 3 \end{vmatrix} \end{bmatrix} :$$

$$= 1 + \text{rank} \begin{bmatrix} -2-0 \\ -1-15 \\ 3+6 \end{bmatrix}$$

$$= 1 + \text{rank} \begin{bmatrix} -2 \\ -16 \\ 9 \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} \begin{vmatrix} -2 & 0 \\ -16 & 0 \end{vmatrix} \\ \begin{vmatrix} -2 & 0 \\ 9 & 0 \end{vmatrix} \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{rank } A = 2$$

(ii)
$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 0 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix}$$

Sol.

Let $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 0 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix}$

Then

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$$\text{rank } A = 1 + \text{rank} \begin{bmatrix} \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ 2 & 6 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ -2 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ -2 & 3 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ -1 & 4 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ -1 & -2 \end{vmatrix} \end{bmatrix}$$

$$= 1 + \text{rank} \begin{bmatrix} -3 & 6 \\ 3 & -3 \\ 6 & -5 \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} \begin{vmatrix} -3 & 6 \\ 3 & -3 \end{vmatrix} \\ \begin{vmatrix} -3 & 6 \\ 6 & -5 \end{vmatrix} \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} 9-18 \\ 15-36 \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} -9 \\ -21 \end{bmatrix}$$

$$= 3 + \text{rank} \begin{bmatrix} -9 & 0 \\ -21 & 6 \end{bmatrix}$$

$$= 3 + \text{rank} \begin{bmatrix} 0 \end{bmatrix}$$

$$= 3$$

(iii)

$$\begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$$

Sol.

$$\text{Let } A = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$$

Then

$$\text{rank } A = 1 + \text{rank} \begin{bmatrix} \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ 1 & -4 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & -4 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 2 & -7 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ 2 & -3 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ 3 & 8 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 3 & -7 \end{vmatrix} & \begin{vmatrix} 1 & -3 \\ 3 & -8 \end{vmatrix} \end{bmatrix}$$

$$= 1 + \text{rank} \begin{bmatrix} 1 & 2 & 1 & -1 \\ -3 & -6 & -3 & 3 \\ -1 & -2 & -1 & 1 \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} \begin{vmatrix} 1 & 2 \\ -3 & -6 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -3 & -3 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ -3 & 3 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} \end{bmatrix}$$

$$= 2 + \text{rank} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 2 + 0$$

$$= 2$$

(iv)

$$\begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 1 & 4 & 1 & 3 & 5 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix}$$

Sol.

$$\text{Let } A = \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 1 & 4 & 1 & 3 & 5 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix}$$

then

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$$\text{rank } A = 1 + \text{rank} \left[\begin{array}{cc|cc} 1 & 3 & 1 & -2 \\ 1 & 4 & 1 & -1 \\ 1 & 3 & 1 & -2 \\ 2 & 7 & 2 & -3 \end{array} \right]$$

$$= 1 + \text{rank} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 1 & 4 & -1 & -1 \\ 1 & 1 & -4 & 5 \end{array} \right]$$

$$= 2 + \text{rank} \left[\begin{array}{cc|cc} 1 & 3 & 1 & -2 \\ 1 & 4 & 1 & -1 \\ 1 & 3 & 1 & -2 \\ 1 & 1 & 1 & -4 \end{array} \right]$$

$$= 2 + \text{rank} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ -2 & -2 & 4 & -1 \end{array} \right]$$

$$= 3 + \text{rank} \left[\begin{array}{cc|cc} 1 & 1 & 1 & -2 \\ -2 & -2 & -2 & 4 \end{array} \right]$$

$$= 3 + \text{rank} \left[\begin{array}{cc} 0 & 0 \end{array} \right]$$

$$= 3 + 0$$

$$\text{rank } A = 3$$



Exercise No. 6.4

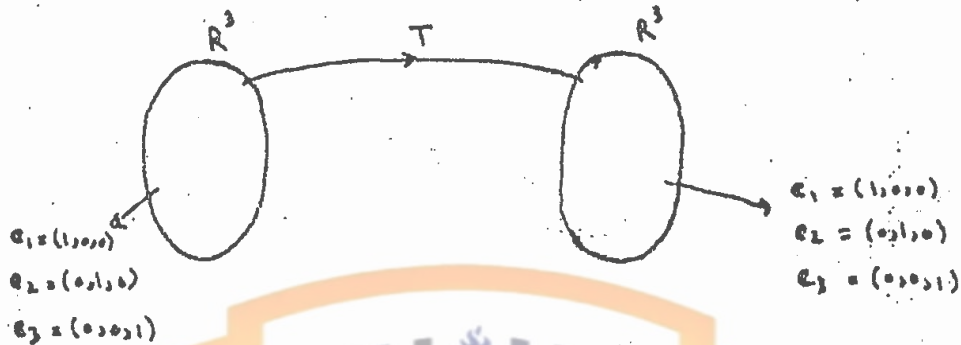


Q1 Find the matrix of each of the following linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$ with respect to the standard basis for $\mathbb{R}^3$:

(i) $T(x_1, x_2, x_3) = (x_1, x_2, 0)$

Sol. Given linear transformation is

$$T(x_1, x_2, x_3) = (x_1, x_2, 0)$$



Now

$$T(1, 0, 0) = (1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1)$$

$$T(0, 1, 0) = (0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$$

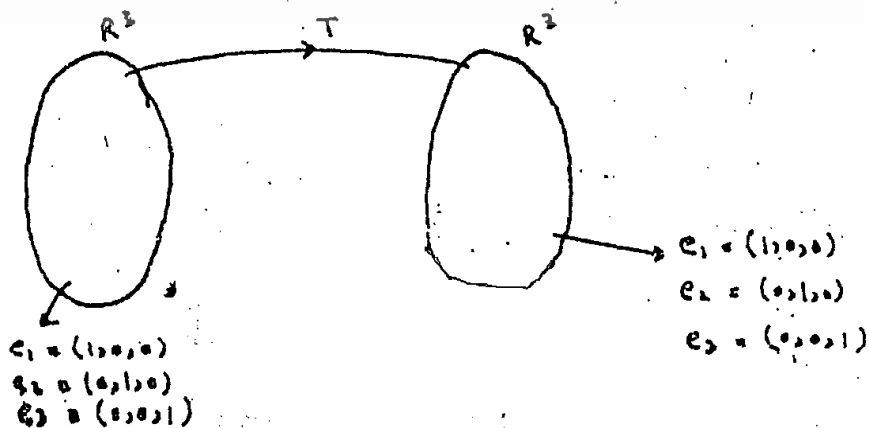
$$T(0, 0, 1) = (0, 0, 0) = 0(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1)$$

Hence matrix of linear transformation T is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(ii) $T(x_1, x_2, x_3) = (x_1 + x_2, -x_1 - x_2, x_3)$

Sol.



Here $T(x_1, x_2, x_3) = (x_1 + x_2, -x_1 - x_2, x_3)$

Then $T(1, 0, 0) = (1, -1, 0) = 1(1, 0, 0) + (-1)(0, 1, 0) + 0(0, 0, 1)$

$T(0, 1, 0) = (1, -1, 0) = 1(1, 0, 0) + (-1)(0, 1, 0) + 0(0, 0, 1)$

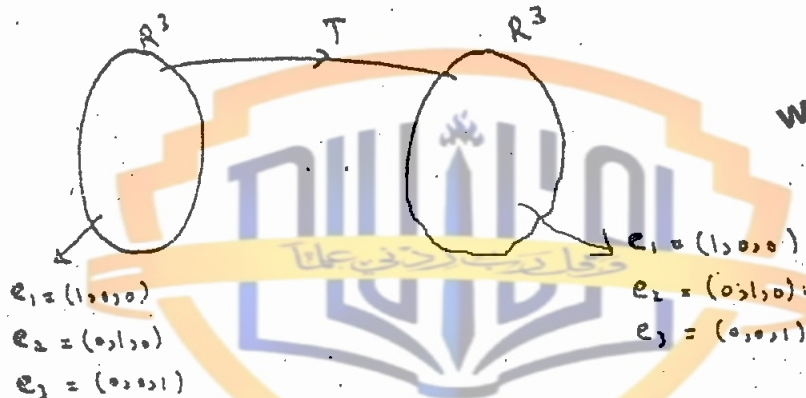
$T(0, 0, 1) = (0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$

Hence matrix of linear transformation T is

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(iii) $T(x_1, x_2, x_3) = (x_2, -x_1, -x_3)$

Sol.



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Here $T(x_1, x_2, x_3) = (x_2, -x_1, -x_3)$

Then $T(1, 0, 0) = (0, -1, 0) = 0(1, 0, 0) + (-1)(0, 1, 0) + 0(0, 0, 1)$

$T(0, 1, 0) = (1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1)$

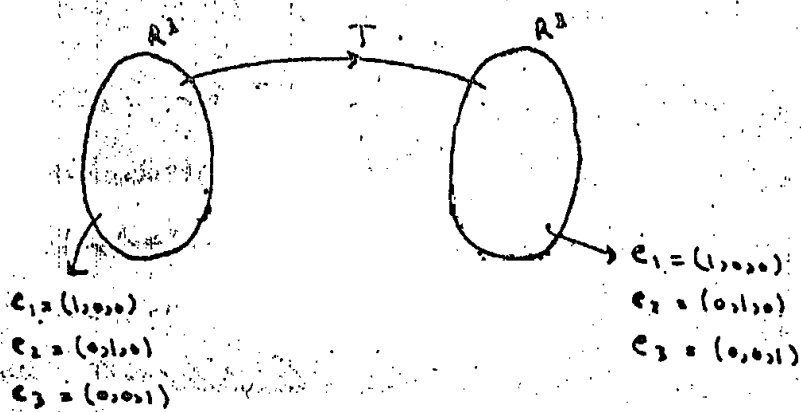
$T(0, 0, 1) = (0, 0, -1) = 0(1, 0, 0) + 0(0, 1, 0) + (-1)(0, 0, 1)$

Hence matrix of linear transformation T is

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(iv) $T(x_1, x_2, x_3) = (x_1, x_2 + x_3, x_1 + x_2 + x_3)$

Sol.



Here $T(x_1, x_2, x_3) = (x_1, x_2 + x_3, x_1 + x_2 + x_3)$

Then $T(1, 0, 0) = (1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$

$T(0, 1, 0) = (0, 1, 1) = 0(1, 0, 0) + 1(0, 1, 0) + 1(0, 0, 1)$

$T(0, 0, 1) = (0, 1, 1) = 0(1, 0, 0) + 1(0, 1, 0) + 1(0, 0, 1)$

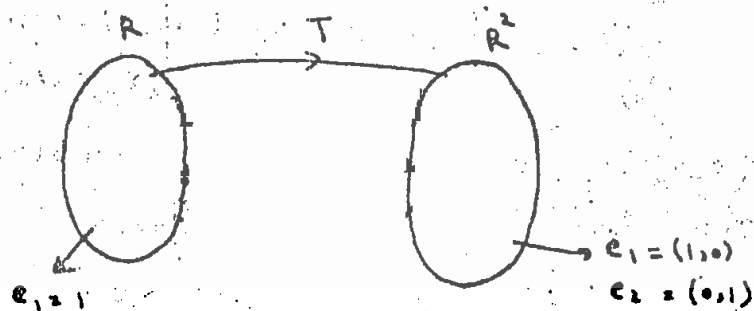
Hence matrix of linear transformation T is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q2 Find the matrix of each of the following linear transformations with respect to the standard bases of the given spaces:

(i) $T: \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $T(x) = (3x, 5x)$

Sol.



Here $T(x) = (3x, 5x)$

then $T(1) = (3, 5) = 3(1, 0) + 5(0, 1)$

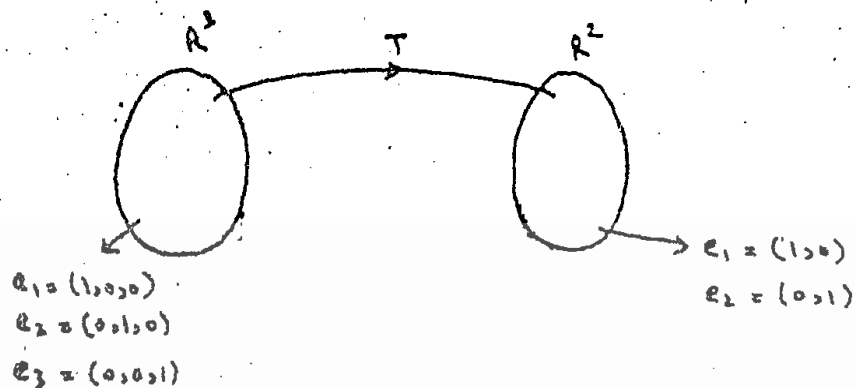
Hence matrix of linear transformation T is

$$\begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

(ii) $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ defined by

$$T(x_1, x_2, x_3) = (3x_1 - 4x_2 + 9x_3, 5x_1 + 3x_2 - 2x_3)$$

Sol.



Here, $T(x_1, x_2, x_3) = (3x_1 - 4x_2 + 9x_3, 5x_1 + 3x_2 - 2x_3)$

Then $T(1, 0, 0) = (3, 5) = 3(1, 0) + 5(0, 1)$

$T(0, 1, 0) = (-4, 3) = -4(1, 0) + 3(0, 1)$

$T(0, 0, 1) = (9, -5) = 9(1, 0) + (-5)(0, 1)$

Hence matrix of linear transformation T is

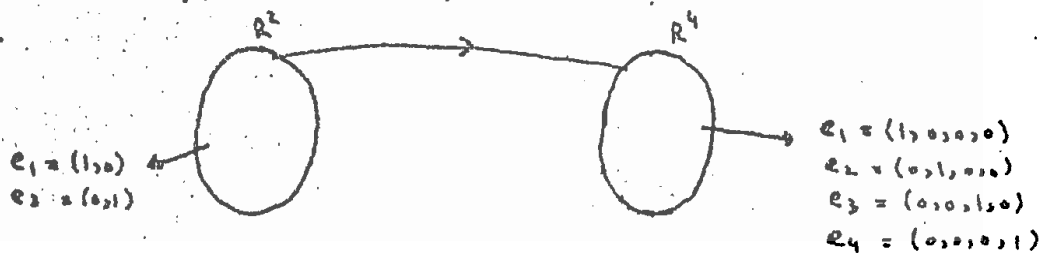
$$\begin{bmatrix} 3 & -4 & 9 \\ 5 & 3 & -5 \end{bmatrix}$$

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(iii) $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^4$ defined by

$$T(x_1, x_2) = (3x_1 + 4x_2, 5x_1 - 2x_2, x_1 + 7x_2, 4x_1)$$

Sol.



Here $T(x_1, x_2) = (3x_1 + 4x_2, 5x_1 - 2x_2, x_1 + 7x_2, 4x_1)$

Then $T(1, 0) = (3, 5, 1, 4) = 3(1, 0, 0, 0) + 5(0, 1, 0, 0) + 1(0, 0, 1, 0) + 4(0, 0, 0, 1)$

$T(0, 1) = (4, -2, 7, 0) = 4(1, 0, 0, 0) - 2(0, 1, 0, 0) + 7(0, 0, 1, 0) + 0(0, 0, 0, 1)$

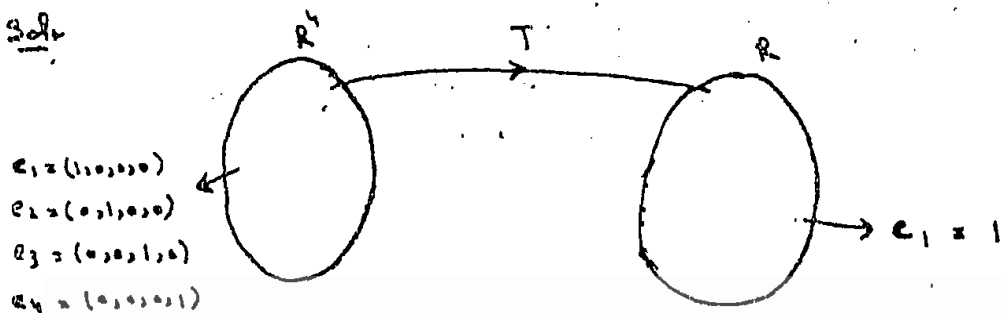
Hence matrix of linear transformation T is

$$\begin{bmatrix} 3 & 4 \\ 5 & -2 \\ 1 & 7 \\ 4 & 0 \end{bmatrix}$$

(iv) $T: \mathbb{R}^4 \rightarrow \mathbb{R}$ defined by

$$T(x_1, x_2, x_3, x_4) = 2x_1 + 3x_2 - 7x_3 + x_4$$

Sol.



Here $T(x_1, x_2, x_3, x_4) = 2x_1 + 3x_2 - 7x_3 + x_4$

Then $T(1, 0, 0, 0) = 2 = 2 \cdot 1$

$T(0, 1, 0, 0) = 3 = 3 \cdot 1$

$T(0, 0, 1, 0) = -7 = -7 \cdot 1$

$T(0, 0, 0, 1) = 1 = 1 \cdot 1$

Hence matrix of linear transformation T is

$$\begin{bmatrix} 2 & 3 & -7 & 1 \end{bmatrix}$$

Q3 Each of the following is the matrix of a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$. Determine m, n & express T in terms of Co-ordinates.

(i)

$$\begin{bmatrix} 3 & 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 1 & 1 \end{bmatrix}$$

Sol. Given linear transformation is

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Here $n = \text{no. of columns} = 5$

& $m = \text{no. of rows} = 3$

Now

let $x = (x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$

then T is defined as

$$= \begin{bmatrix} 3 & 1 & 0 & 2 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 3x_1 + x_2 + 0x_3 + 2x_4 + x_5 \\ x_1 + 0x_2 + 0x_3 + x_4 + x_5 \\ 0x_1 - x_2 + x_3 + x_4 + x_5 \end{bmatrix}$$

or

$$T(x_1, x_2, x_3, x_4, x_5) = (3x_1 + x_2 + 2x_4 + x_5, x_1 + x_4 + x_5, -x_2 + x_3 + x_4 + x_5)$$

which is T in terms of Co-ordinates.

(ii)

$$\begin{bmatrix} 6 & -1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

Sol. Given linear transformation is

$$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

Here $n = \text{no. of columns} = 2$

+ $m = \text{no. of rows} = 3$

Now let $x = (x_1, x_2) \in \mathbb{R}^2$

then T is defined as

$$T(x) = Ax$$

$$= \begin{bmatrix} 6 & -1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{or } T(x) = \begin{bmatrix} 6x_1 - x_2 \\ x_1 + 2x_2 \\ x_1 + 3x_2 \end{bmatrix}$$

or $T(x_1, x_2) = (6x_1 - x_2, x_1 + 2x_2, x_1 + 3x_2)$
 which is T in terms of Co-ords.

(iii) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 5 & 6 \\ -2 & 3 & -1 \end{bmatrix}$

Sol. Given linear transformation is

$$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

Here $n = \text{no. of Columns} = 3$

& $m = \text{no. of rows} = 3$

Let $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ then

T is defined as

$$T(x) = Ax$$

$$= \begin{bmatrix} 1 & 1 & 2 \\ 2 & 5 & 6 \\ -2 & 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} x_1 + x_2 + 2x_3 \\ 2x_1 + 5x_2 + 6x_3 \\ -2x_1 + 3x_2 - x_3 \end{bmatrix}$$

or

$$T(x_1, x_2, x_3) = (x_1 + x_2 + 2x_3, 2x_1 + 5x_2 + 6x_3, -2x_1 + 3x_2 - x_3)$$

which is T in terms of Co-ords.

Q4 The matrix of a linear transformation

$T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

Find T in terms of Co-ords.

& its matrix w.r.t. the basis

$$v_1 = (0, 1, 2), v_2 = (1, 1, 1), v_3 = (1, 0, -2).$$

Sol. Given linear transformation is

$$T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

Let $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ then T is defined as

$$T(x) = Ax$$

or

$$= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 0x_1 + x_2 + x_3 \\ x_1 + 0x_2 - x_3 \\ -x_1 - x_2 + 0x_3 \end{bmatrix}$$

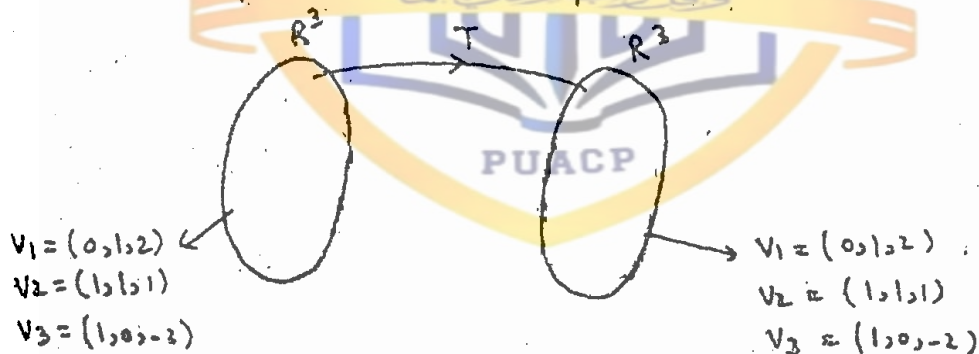
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or

$$T(x_1, x_2, x_3) = (x_2 + x_3, x_1 - x_3, -x_1 - x_2)$$

which is T in terms of co-ords.

Now we find matrix of T



$$\text{Here } T(x_1, x_2, x_3) = (x_2 + x_3, x_1 - x_3, -x_1 - x_2)$$

$$\text{Then } T(0, 1, 2) = (3, -2, -1) = \alpha(0, 1, 2) + \beta(1, 1, 1) + \gamma(1, 0, -2)$$

$$(3, -2, -1) = (b + c, a + b, 2a + b - 2c)$$

$$\Rightarrow b + c = 3 \quad \text{--- ①}$$

$$a + b = -2 \quad \text{--- ②}$$

$$2a + b - 2c = -1 \quad \text{--- ③}$$

$$\text{①} - \text{②} \Rightarrow c - a = 5 \quad \text{--- ④}$$

$$\text{②} - \text{③} \Rightarrow -a + 2c = -1 \quad \text{--- ⑤}$$

Subst. ④ from ⑤

$$-C = 6$$

$$\text{or } \boxed{C = -6}$$

Put in ①

$$b - 6 = 2$$

$$\text{or } \boxed{b = 8}$$

Put in ②

$$a + 9 = -2$$

$$\boxed{a = -11}$$

Now

$$T(1,1,1) = (2,0,-2) = \overset{-2}{a}(0,1,2) + \overset{2}{b}(1,1,1) + \overset{0}{c}(1,0,-2) \\ = (b+c, a+b, 2a+b-2c)$$

$$\Rightarrow b+c = 2 \quad \text{--- ①}$$

$$a+b = 0 \quad \text{--- ②}$$

$$2a+b-2c = -2 \quad \text{--- ③}$$

$$\text{①} - \text{②} \Rightarrow C - a = 2$$

$$-a + 2C = 2$$

Put. we get

$$-C = 0 \Rightarrow \boxed{C = 0}$$

Put in ①

$$b + 0 = 2 \Rightarrow \boxed{b = 2}$$

Put in ②

$$a + 2 = 0 \Rightarrow \boxed{a = -2}$$

Now

$$T(1,0,-2) = (-2,3,-1) = \overset{14}{a}(0,1,2) + \overset{-11}{b}(1,1,1) + \overset{9}{c}(1,0,-2) \\ = (b+c, a+b, 2a+b-2c)$$

$$\Rightarrow b+c = -2 \quad \text{--- ①}$$

$$a+b = 3 \quad \text{--- ②}$$

$$2a+b-2c = -1 \quad \text{--- ③}$$

$$\text{①} - \text{②} \Rightarrow$$

Adding ① & ②

$$2a = 0 \Rightarrow \boxed{a = 0}$$

Put in ①

$$0 - b = 0 \Rightarrow \boxed{b = 0}$$

Hence vectors $(1,1)$ & $(-1,1)$ are linearly independent & since they are two in number, so they form a basis for $\mathbb{R}^2$.

Let (x_1, x_2) be an arbitrary vector of $\mathbb{R}^2$
Then (x_1, x_2) can be expressed as

$$\begin{aligned}(x_1, x_2) &= a(1,1) + b(-1,1) \\ &= (a-b, a+b)\end{aligned}$$

 $\Rightarrow$

$$\begin{array}{rcl} a-b & = & x_1 \quad \text{--- ①} \\ a+b & = & x_2 \quad \text{--- ②} \end{array}$$

$$2a = x_1 + x_2$$

$$\text{or } \boxed{a = \frac{x_1 + x_2}{2}}$$

Put in ①

$$\frac{x_1 + x_2}{2} - b = x_1$$

$$b = \frac{x_1 + x_2}{2} - x_1$$

$$= \frac{x_1 + x_2 - 2x_1}{2}$$

$$\boxed{b = \frac{x_2 - x_1}{2}}$$

$$\text{So } (x_1, x_2) = \left(\frac{x_1 + x_2}{2}\right)(1,1) + \left(\frac{x_2 - x_1}{2}\right)(-1,1)$$

Applying T on both sides

$$T(x_1, x_2) = T\left\{\left(\frac{x_1 + x_2}{2}\right)(1,1) + \left(\frac{x_2 - x_1}{2}\right)(-1,1)\right\}$$

$$= \left(\frac{x_1 + x_2}{2}\right)T(1,1) + \left(\frac{x_2 - x_1}{2}\right)T(-1,1)$$

 $\therefore T$ is linear

$$c - a = -5$$

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$$\textcircled{1} - \textcircled{3} \Rightarrow -a + 2c = 4$$

Soln. we get

$$-c = -9 \Rightarrow \boxed{c = 9}$$

Put in $\textcircled{1}$

$$b + 9 = -2 \Rightarrow \boxed{b = -11}$$

Put in $\textcircled{2}$

$$a - 11 = 3 \Rightarrow \boxed{a = 14}$$

Hence, matrix of linear transformation T

$$\begin{bmatrix} -11 & -2 & 14 \\ 9 & 2 & -11 \\ -6 & 0 & 9 \end{bmatrix}$$

Q5 A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ maps the vector $(1,1)$ into $(0,1,2)$ & the vector $(-1,1)$ into $(2,1,0)$. What matrix does T represent with respect to the standard bases for $\mathbb{R}^2$ & $\mathbb{R}^3$?

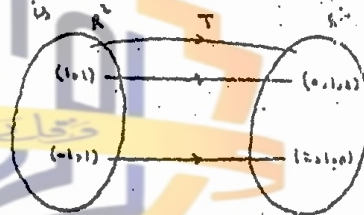
Soln. Given linear transformation is

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

We are given that

$$T(1,1) = (0,1,2)$$

$$\& T(-1,1) = (2,1,0)$$



First we check linear independency of vectors $(1,1)$ & $(-1,1)$

suppose for scalars a, b

$$a(1,1) + b(-1,1) = 0$$

$$(a-b, a+b) = (0,0)$$

$$\Rightarrow a-b = 0 \quad \text{--- (1)}$$

$$a+b = 0 \quad \text{--- (2)}$$

$$T(x_1, x_2) = \left(\frac{x_1+x_2}{2}\right)(0,1,2) + \left(\frac{x_1-x_2}{2}\right)(2,1,0)$$

$$= \left(x_2 - x_1, \frac{x_1+x_2}{2} + \frac{x_2-x_1}{2}, x_1+x_2\right)$$

$$\text{or } T(x_1, x_2) = (x_2 - x_1, x_2, x_1 + x_2)$$

which is T in terms of co-ords.

$$\text{Now or } T(x_1, x_2) = (x_2 - x_1, x_2, x_1 + x_2)$$

$$\text{then } T(1,0) = (-1,0,1)$$

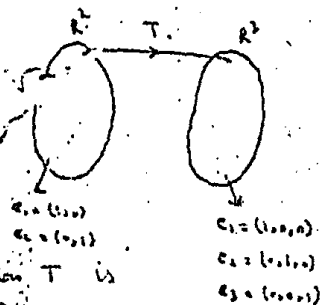
$$= -1(1,0,0) + 0(0,1,0) + 1(0,0,1)$$

$$\& T(0,1) = (1,1,1)$$

$$= 1(1,0,0) + 1(0,1,0) + 1(0,0,1)$$

Hence matrix of linear transformation T is

$$\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$



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